More study time is only part of the story.

Message: Ask what help a student needs before prescribing longer study hours. Study-time groups differ on average, but time alone explains little grade variation. Earlier failures should prompt supportive follow-up, never a fixed label of ability.

Question and data. We tested whether mean final Portuguese grade (G3, 0–20) differs across four recorded weekly study-time bands: <2, 2–5, 5–10 and >10 hours. The original UCI file has 649 students at two Portuguese schools, 33 columns, no missing values and 15 zero grades retained. It is an observational sample, not a recent Korean survey.

Analysis	Result	Meaning
Primary categorical study time	$F(3,645) = 15.88$; $p < 0.001$; $R^2 = 6.88\%$	Mean grades differ, but the model explains a small share of variation.
Highest minus lowest band	+2.21 points; 95% CI [1.09, 3.33]	An average difference, not an individual prediction.
Numeric-code simple regression	$G3 = 10.028 + 0.973 \times$ code; $R^2 = 6.24\%$	A code step is not an additional hour.
Past failures / family support	$R^2 = 19.29\%$ / 0.35%; family-support $p = 0.132$	Separate models; exploratory associations.

Prediction. On the same five held-out folds, study-time categorical regression has $R^2 = 5.09\%$ and RMSE = 3.14 points; a 30-predictor random forest has $R^2 = 28.22\%$ and RMSE = 2.74. G1/G2 are excluded. These held-out results must be distinguished from the in-sample figures above. Prediction does not identify causes.

Three perspectives. Map A reports measured associations. Map B proposes motivation, prior mastery and learning gained as possible mechanisms. Map C evaluates held-out prediction. The non-significant yes/no family-support item may miss meaningful differences in support quality.

Proposed action. A teacher or counselor can invite a private, optional conversation, check mastery and study methods, agree on a two-week support plan, and review learning progress and burden. This protocol is a proposal, not an evaluated intervention. A consented comparison design is needed to assess causal improvement.

Limits and next steps. Unequal time bands, a small >10-hour group (n = 35), unmeasured factors, exploratory screening and restricted school coverage limit interpretation. No optimal study-time range is established. Collect recent local data before applying conclusions to Korean schools.

Final improvements. Plain-language takeaways, a concrete support protocol, presentation scenes that reset to all 649 records, and a separate browser-local CSV workspace respond to peer feedback. Uploaded analyses do not change the original results.

Source: Cortez, P. (2008). Student Performance. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T (CC BY 4.0). Original data, executed notebook and website source accompany this submission.